Shiyao Qian, B.A.Sc

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Research Interests

3D Generation · 3D Reconstruction · 3D Gaussian Splatting

Education

2022 – 2027 **B.A.Sc, University of Toronto** Computer Engineering

- GPA: *3.85/4.0*
- Relevant Coursework: Applied Fundamentals of Deep Learning, Algorithms & Data Structures, Introduction to Artificial Intelligence, Probability & Applications, Applied Linear Algebra

Research Experience

3D Vision Foundation Models with Spatial Geometric Priors
Research Intern, Spatial Intelligence Group, Huawei Noah's Ark Lab
Supervisor: Yuan Ren Jan. 2026 – Apr. 2026

- Investigated injection of gravity direction and residual optical flow as privileged information to fine-tune feedforward 3D foundation models (VGGT, PI3) for dynamic-scene generalization; experiments validated the effectiveness of gravity priors and optical flow in improving fine-tuning performance on scene reconstruction benchmarks
- Built data preprocessing pipelines for both priors: gravity direction extracted via GeoCalib from images and camera intrinsics across 10 datasets; dense optical flow predicted by RAFT, with the dominant ego-motion component approximated by RANSAC-fitting a 2×3 affine transformation to the flow field. Extended the pipeline post-submission to 3D residual scene flow, obtained by lifting optical flow with forward disparity and subtracting camera-induced motion across 5 dynamic datasets.

Part-Level 3D Gaussian Vehicle Generation with Joint and Hinge Axis Estimation
Research Intern, Spatial Intelligence Group, Huawei Noah's Ark Lab
Supervisor: Yuan Ren May 2025 – Nov. 2025

- Led development of a 3D Gaussian Splatting–based part-level articulated vehicle generation framework; filed one Huawei patent application
- Designed a three-stage pipeline: (1) SAM2- and YOLO-guided multi-view mask extraction with contrastive per-Gaussian feature learning for part clustering, (2) dual PointNet++ networks for joint part segmentation and kinematic parameter prediction over Gaussian-derived point clouds, and (3) mask-guided geometric refinement of Gaussian primitives for clean part boundaries
- Constructed a custom point cloud dataset with part annotations rigorously derived from native CAD 3D attributes rather than 2D-mask projection, providing accurate part attribution and clean part boundaries absent from existing public 3D datasets
- Validated downstream controllability on CARLA's articulated vehicle assets, achieving CLIP similarity of 0.873 and LPIPS of 0.283; ablation studies confirmed the contribution of the fine-tuning module and YOLO mask integration to perceptual fidelity

Research Experience (continued)

Mitigating Age-Related Bias in Large Language Models
Research Assistant — Feb. 2024 – May 2025
Supervisor: Prof. Zhuang Liu (Associate Professor, Dongbei University of Finance and Economics)

- Co-authored peer-reviewed publication in INFORMS Journal on Computing (2025) on age-related bias mitigation in LLMs; contributed framework design, experimental execution, dataset construction, and manuscript writing as second author
- Designed the BMIL (Bias Mitigation in the Loop) parameter-free debiasing framework and the Coop-BMIL multi-agent debate system, using reinforcement learning for collaborative bias resolution across LLaMA-family models

SCORE: Story Coherence and Retrieval Enhancement for AI Narratives
Research Assistant
Supervisor: Tianyu Shi (Ph.D., University of Toronto) — Aug. 2024 – Jan. 2025

- Contributed to SCORE, an AI-prompted framework for narrative coherence in LLM-generated stories; integrated with Retrieval-Augmented Generation (RAG) for question-answering applications
- Fine-tuned T5 on the WritingPrompts dataset; constructed multi-model evaluation set across GPT and Gemini outputs

Research Publications

1. **S. Qian**, Y. Ren, D. Bai, and B. Liu, "Part-Level 3D Gaussian Vehicle Generation with Joint and Hinge Axis Estimation," Submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026, 2026. arXiv: 2604.05070 [cs.AI]. URL: https://arxiv.org/abs/2604.05070
2. Z. Liu, **S. Qian**, S. Cao, and T. Shi, "Mitigating age-related bias in large language models: Strategies for responsible artificial intelligence development," *INFORMS Journal on Computing*, 2025. URL: https://pubsonline.informs.org/doi/10.1287/ijoc.2024.0645
3. Q. Yi, Y. He, J. Wang, X. Song, K. Lu, **S. Qian**, et al., "Score: Story coherence and retrieval enhancement for ai narratives," in *Companion Proceedings of the ACM Web Conference 2026*, 2026, pp. 823–827. URL: https://dl.acm.org/doi/abs/10.1145/3774905.3795734

Skills

Deep Learning	PyTorch, Hydra, Huggingface, Tensorboard
Systems and Infrastructure	Bash, Tmux, Slurm, Git
Computer Vision Models	VGGT, PI3, AnySplat, SAGA, TRELLIS, SAM2, YOLO, RAFT, GeoCalib, PointNet++
3D Graphics & Geometry	Point Cloud, Mesh, 3D Gaussian Splatting, gsplat, Colmap, Open3D, Blender, CARLA Simulator
Programming Languages	Python, C++

Awards and Honors

2023	**John M. Empey Scholarship**, highest academic achievement in first-year written exams
2022-2025	**Dean's Honour List**, average over 80 every semester since enrollment